



Furnace Pressure Switch Testing

80% and 90% furnaces | Field Standard Operating Procedure | Brackett Heating, Air & Plumbing

BEFORE YOU START

The pressure switch is a safety that proves the furnace can vent. Most of the time a pressure switch fault is the switch reporting a real venting problem, not the switch failing.

Do not leave a pressure switch jumpered. A bypassed switch can vent combustion products into an occupied building. Wear a personal CO monitor.

1. What the switch is actually doing

The pressure switch proves the inducer is moving combustion products before the board will allow ignition. It is normally open and closes when the inducer develops enough negative pressure. It is not a component that fails often.

	80% (induced draft)	90% (condensing)
Switches present	Usually one draft-proving switch	Two or more: draft proving plus blocked-drain; two-stage units add a second set point
Senses	Negative pressure at the inducer housing tap	Negative pressure at the inducer or collector box; some sealed-combustion designs use a positive-pressure switch
Typical set point	Commonly -0.40 to -0.70 in. W.C.	Commonly -0.25 to -1.20 in. W.C. depending on stage and model
Most common real cause	Restricted or disconnected vent, plugged inducer tap, weak inducer	Plugged condensate drain or trap - check this first

The set point is printed on the switch body, for example "-0.60 IN W.C." That is the pressure at which the contacts close. Replace only with the same set point and the same configuration (SPST normally open, or SPDT where the board uses both contacts).

2. Tools

- Digital manometer with 0.01 in. W.C. resolution, or a low-range inclined manometer.
- Multimeter, set to VAC for in-circuit testing.
- Tee fitting and spare hose to splice into the switch port.
- Soft wire or a drill bit turned by hand to clear a plugged pressure tap.

3. Read the fault code first

- **Pressure switch open** (did not close): the inducer ran but the switch never made. Work through sections 4 and 5.
- **Pressure switch closed** (stuck closed): the board saw the switch already made before the inducer started. Usually a welded switch, a shorted wire, or a board miswire. Confirm with the electrical test in section 4 before the inducer starts - any voltage reading other than 24 V across an at-rest switch means the contacts are made when they should be open.

4. Electrical test - is the switch making?

Done live, in circuit, with the furnace running. It tells you whether the contacts are closing without removing anything.

- 1 Set the meter to VAC.
- 2 Call for heat and let the inducer come up to speed.
- 3 Place the meter leads across the two switch terminals.
- 4 **0 volts across the switch** means the contacts are closed and passing. The switch is doing its job.
- 5 **24 volts across the switch** means the contacts are open. Either the switch is bad or it is not getting the pressure signal - go to section 5 to find out which.

DO NOT LEAVE A SWITCH JUMPERED

Bypassing a pressure switch to confirm a diagnosis is acceptable only momentarily, with the technician present and a CO monitor running. Remove the jumper immediately.

5. Pressure test - is the signal there?

This is the test that separates a bad switch from a venting problem, and it is the step most often skipped.

- 1 Tee the manometer into the hose at the switch port, or connect directly to the inducer tap.
- 2 Call for heat and let the inducer run.
- 3 Read the actual negative pressure developed and compare it to the set point printed on the switch.
- 4 **Pressure meets or exceeds the set point but the switch does not close:** the switch has failed. Replace it with the identical set point.
- 5 **Pressure does not reach the set point:** the switch is fine. The fault is upstream - work through section 6 or 7.

6. Common causes on 80% furnaces

- Restricted, disconnected, or blocked flue - bird nest, debris, collapsed liner, or a disconnected vent connector.
- Plugged pressure tap in the inducer housing. Soot and rust close these up. Clear gently with soft wire or a drill bit turned by hand.
- Cracked, softened, collapsed, or disconnected hose. Hoses go brittle with heat age - replace rather than refit.
- Weak inducer: failing bearings, a wheel loose or corroded on the shaft, or a motor down on speed.
- Sooted heat exchanger restricting flow through the passages.
- Blocked combustion air supply to the space.
- Improper vent sizing or an excessive number of elbows on the connector.

7. Common causes on 90% furnaces

- **Plugged condensate drain or a full trap.** Check this first on any condensing furnace - it is the single most common cause.
- Water standing in the pressure hose, the collector box, or the inducer housing.
- Blocked or restricted PVC intake or exhaust: ice at the termination, insects, insulation, snow, or a screen somebody added.
- Termination too close to grade or drifted shut.
- Vent run exceeding the maximum equivalent length for the pipe diameter and elbow count.
- Kinked, collapsed, or cracked hose between the tap and the switch.
- Plugged drain port in the inducer housing or collector box.
- Cracked collector box or a failed secondary heat exchanger.

8. Two-stage and high-altitude notes

- Two-stage condensing furnaces use separate set points for low and high fire, for example -0.35 in. W.C. low and -0.85 in. W.C. high. Test each switch at its own stage, not at whatever stage the furnace happens to be in.
- High-altitude kits include different pressure switches with lower set points. A switch that looks wrong may simply be the correct altitude part - check the kit documentation before condemning it.
- When replacing, match the set point exactly. A switch with a lower set point will mask a venting fault.

9. Verify the repair

- 1 Run a full heat cycle from a cold start.
- 2 Confirm the switch closes within a few seconds of the inducer reaching speed.
- 3 Confirm it reopens at the end of the cycle.
- 4 Re-read the negative pressure and record it against the set point.
- 5 Run a combustion analysis and record CO. A venting fault that was tripping the switch may have been making CO.
- 6 Cycle the furnace twice more to be sure the fault does not return intermittently.

THE RULE OF THUMB

If the pressure signal is there and the switch will not close, replace the switch. If the pressure signal is not there, the switch is telling you the truth - find out why the furnace cannot vent.